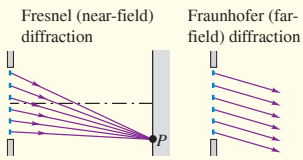
CHAPTER 36 SUMMARY

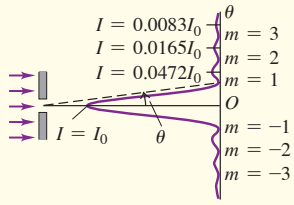
Fresnel and Fraunhofer diffraction: Diffraction occurs when light passes through an aperture or around an edge. When the source and the observer are so far away from the obstructing surface that the outgoing rays can be considered parallel, it is called Fraunhofer diffraction. When the source or the observer is relatively close to the obstructing surface, it is Fresnel diffraction.



Single-slit diffraction: Monochromatic light sent through a narrow slit of width a produces a diffraction pattern on a distant screen. Equation (36.2) gives the condition for destructive interference (a dark fringe) at a point P in the pattern at angle θ. Equation (36.7) gives the intensity in the pattern as a function of θ. (See Examples 36.1–36.3.)

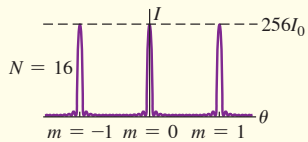
$$\sin\theta = \frac{m\lambda}{a} \quad (m = \pm 1, \pm 2, \pm 3, \ldots) \tag{36.2}$$

$$I = I_0\left\{\frac{\sin[\pi a(\sin\theta)/\lambda]}{\pi a(\sin\theta)/\lambda}\right\}^2 \tag{36.7}$$



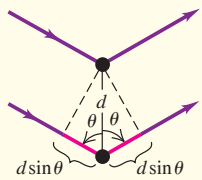
Diffraction gratings: A diffraction grating consists of a large number of thin parallel slits, spaced a distance d apart. The condition for maximum intensity in the interference pattern is the same as for the two-source pattern, but the maxima for the grating are very sharp and narrow. (See Example 36.4.)

$$d\sin\theta = m\lambda \quad (m = 0, \pm 1, \pm 2, \pm 3, \ldots) \tag{36.13}$$



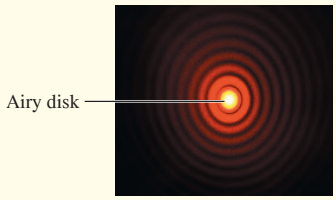
X-ray diffraction: A crystal serves as a three-dimensional diffraction grating for x rays with wavelengths of the same order of magnitude as the spacing between atoms in the crystal. For a set of crystal planes spaced a distance d apart, constructive interference occurs when the angles of incidence and scattering (measured from the crystal planes) are equal and when the Bragg condition [Eq. (36.16)] is satisfied. (See Example 36.5.)

$$2d\sin\theta = m\lambda \quad (m = 1, 2, 3, \ldots) \tag{36.16}$$



Circular apertures and resolving power: The diffraction pattern from a circular aperture of diameter D consists of a central bright spot, called the Airy disk, and a series of concentric dark and bright rings. Equation (36.17) gives the angular radius θ_1 of the first dark ring, equal to the angular size of the Airy disk. Diffraction sets the ultimate limit on resolution (image sharpness) of optical instruments. According to Rayleigh's criterion, two point objects are just barely resolved when their angular separation θ is given by Eq. (36.17). (See Example 36.6.)

$$\sin\theta_1 = 1.22\frac{\lambda}{D} \tag{36.17}$$



Chapter 36 Media Assets

For assigned homework and other learning materials, go to Mastering Physics.

GUIDED PRACTICE

KEY EXAMPLE VARIATION PROBLEMS

Be sure to review EXAMPLE 36.1 (Section 36.2) and EXAMPLES 36.2 and 36.3 (Section 36.3) before attempting these problems.

VP36.3.1 You pass laser light of wavelength 645 nm through a slit 0.250 mm in width and observe the diffraction pattern on a screen a large distance away. On the screen, the centers of the second minima on either side of the central bright fringe are 28.0 mm apart. (a) How far away is the screen? (b) What would the distance be between these minima if the wavelength were 525 nm?

VP36.3.2 You shine a laser on a narrow slit 0.221 mm in width. In the diffraction pattern that appears on a screen 5.00 m from the slit, the third minimum is 45.7 mm from the middle of the central bright fringe. Find (a) the wavelength of the laser light and (b) the angle of a line from the center of the slit to the second dark fringe on the screen.

VP36.3.3 At a certain point in a single-slit diffraction pattern there is a phase difference of 35.0 radians between wavelets arriving at the point from the two edges of the slit. The slit is 0.250 mm wide, and the light used has wavelength 545 nm. (a) What is the angle of a line from the center of the slit to this point in the diffraction pattern? (b) If the intensity at the center of the diffraction pattern is I_0, what is the intensity at this point?

VP36.3.4 In a single-slit diffraction experiment, the slit width is 0.185 mm and the wavelength of the light used is 576 nm. (a) Find the angle of a line from the center of the slit to the first dark fringe. (b) You might expect the intensity at one-half the angle found in part (a), which is halfway between the middle of the central bright fringe (where the intensity is I_0) and the first dark fringe (where the intensity is zero), to be $I_0/2$. What is the actual intensity at this angle?

Be sure to review EXAMPLE 36.4 (Section 36.5) and EXAMPLE 36.5 (Section 36.6) before attempting these problems.

VP36.5.1 You shine laser light so that it falls normally on a transmission grating with 825 slits per millimeter. The second-order bright region occurs at an angle of 41.0° from the central maximum. Find (a) the wavelength of the light and (b) the angles for the first-order and third-order bright regions.

VP36.5.2 The first-order bright region created by a diffraction grating occurs at an angle of 24.0° from the central maximum. The wavelength of light that falls normally onto the grating is 625 nm. Find (a) the number of slits per mm on the grating and (b) the angles of the second-order bright region and the third-order bright region (if any).

VP36.5.3 In a certain crystal the spacing between crystal planes is 0.165 nm. (a) If you shine a beam of x rays of wavelength 0.124 nm on this crystal, for what angle between the beam and the crystal planes does the first strong interference maximum occur? (b) For what larger angles (if any) do strong interference maxima occur?

VP36.5.4 In an x-ray diffraction experiment there is only one strong interference maximum, and this occurs when the x-ray beam makes an angle of 36.0° with the crystal planes. The spacing between the crystal planes is 0.158 nm. (a) What is the wavelength of the x rays? (b) What would the wavelength have to be to have three strong interference maxima, with the third at an angle of 88.0°?

Be sure to review EXAMPLE 36.6 (Section 36.7) before attempting these problems.

VP36.6.1 You wish to study the radio emission from the sun at wavelength 1.70 cm. In order to see details on the sun's surface no larger than the diameter of the earth, the limit of resolution of the telescope must be 9.00×10^{-5} rad (about 0.005°). Using the Rayleigh criterion, find (a) the minimum diameter your radio telescope must have and (b) the limit of resolution of this telescope at wavelength 21.1 cm.

VP36.6.2 A telephoto lens for a camera has focal length 0.250 m and maximum aperture $f/4.0$. Take the wavelength of visible light to be 5.50×10^{-7} m. (a) Assuming that the resolution is limited by diffraction, how far away can an object be for you to be able to resolve two points on the object that are 5.00 mm apart? (b) How far apart are the corresponding points on the image made by the lens?

VP36.6.3 A distant point source of light (like a star) emits light of wavelength 575 nm. When this light enters a camera whose lens has a focal length of 135 mm, the diffraction pattern formed on the camera's detector has an Airy disk of radius 0.0112 mm. Find (a) the diameter of the lens aperture and (b) the f-number.

VP36.6.4 One important goal of astronomers is to have a telescope in space that can resolve planets like the earth orbiting other stars. If a planet orbits its star at a distance of 1.5×10^{11} m (the radius of the earth's orbit around the sun) and the telescope has a mirror of diameter 8.0 m, how far from the telescope could the star and its planet be if the wavelength used was (a) 690 nm and (b) 1400 nm? Use the Rayleigh criterion and give your answers in light-years (1 ly $= 9.46 \times 10^{15}$ m).

BRIDGING PROBLEM Observing the Expanding Universe

An astronomer who is studying the light from a galaxy has identified the spectrum of hydrogen but finds that the wavelengths are somewhat shifted from those found in the laboratory. In the lab, the H_α line in the hydrogen spectrum has a wavelength of 656.3 nm. The astronomer is using a transmission diffraction grating having 5758 lines/cm in the first order and finds that the first bright fringe for the H_α line occurs at $\pm 23.41°$ from the central spot. How fast is the galaxy moving? Express your answer in m/s and as a percentage of the speed of light. Is the galaxy moving toward us or away from us?

SOLUTION GUIDE

IDENTIFY and SET UP

1. You can use the information about the grating to find the wavelength of the H_α line in the galaxy's spectrum.
2. In Section 16.8 we learned about the Doppler effect for electromagnetic radiation: The frequency that we receive from a moving source, such as the galaxy, is different from the frequency that is emitted. Equation (16.30) relates the emitted frequency, the received frequency, and the velocity of the source (the target variable). The equation $c = f\lambda$ relates the frequency f and wavelength λ through the speed of light c.

EXECUTE

3. Find the wavelength of the H_α spectral line in the received light.
4. Rewrite Eq. (16.30) as a formula for the velocity v of the galaxy in terms of the received wavelength and the wavelength emitted by the source.
5. Solve for v. Express it in m/s and as a percentage of c, and decide whether the galaxy is moving toward us or moving away.

EVALUATE

6. Is your answer consistent with the relative sizes of the received wavelength and the emitted wavelength?

PROBLEMS

•, ••, •••: Difficulty levels. CP: Cumulative problems incorporating material from earlier chapters. CALC: Problems requiring calculus. DATA: Problems involving real data, scientific evidence, experimental design, and/or statistical reasoning. BIO: Biosciences problems.

DISCUSSION QUESTIONS

Q36.1 Why can we readily observe diffraction effects for sound waves and water waves, but not for light? Is this because light travels so much faster than these other waves? Explain.

Q36.2 What is the difference between Fresnel and Fraunhofer diffraction? Are they different *physical* processes? Explain.

Q36.3 You use a lens of diameter D and light of wavelength λ and frequency f to form an image of two closely spaced and distant objects. Which of the following will increase the resolving power? (a) Use a lens with a smaller diameter; (b) use light of higher frequency; (c) use light of longer wavelength. In each case justify your answer.

Q36.4 Light of wavelength λ and frequency f passes through a single slit of width a. The diffraction pattern is observed on a screen a distance x from the slit. Which of the following will *decrease* the width of the central maximum? (a) Decrease the slit width; (b) decrease the frequency f of the light; (c) decrease the wavelength λ of the light; (d) decrease the distance x of the screen from the slit. In each case justify your answer.

Q36.5 In a diffraction experiment with waves of wavelength λ, there will be *no* intensity minima (that is, no dark fringes) if the slit width is small enough. What is the maximum slit width for which this occurs? Explain your answer.

Q36.6 An interference pattern is produced by four parallel and equally spaced narrow slits. By drawing appropriate phasor diagrams, explain why there is an interference minimum when the phase difference ϕ from adjacent slits is (a) $\pi/2$; (b) π; (c) $3\pi/2$. In each case, for which pairs of slits is there totally destructive interference?

Q36.7 **Phasor Diagram for Eight Slits.** An interference pattern is produced by eight equally spaced narrow slits. The caption for Fig. 36.14 claims that minima occur for $\phi = 3\pi/4$, $\pi/4$, $3\pi/2$, and $7\pi/4$. Draw the phasor diagram for each of these four cases, and explain why each diagram proves that there is in fact a minimum. In each case, for which pairs of slits is there totally destructive interference?

Q36.8 A rainbow ordinarily shows a range of colors (see Section 33.4). But if the water droplets that form the rainbow are small enough, the rainbow will appear white. Explain why, using diffraction ideas. How small do you think the raindrops would have to be for this to occur?

Q36.9 Some loudspeaker horns for outdoor concerts (at which the entire audience is seated on the ground) are wider vertically than horizontally. Use diffraction ideas to explain why this is more efficient at spreading the sound uniformly over the audience than either a square speaker horn or a horn that is wider horizontally than vertically. Would this still be the case if the audience were seated at different elevations, as in an amphitheater? Why or why not?

Q36.10 Figure 31.12 (Section 31.2) shows a loudspeaker system. Low-frequency sounds are produced by the *woofer,* which is a speaker with large diameter; the *tweeter,* a speaker with smaller diameter, produces high-frequency sounds. Use diffraction ideas to explain why the tweeter is more effective for distributing high-frequency sounds uniformly over a room than is the woofer.

Q36.11 Information is stored on an audio compact disc, CD-ROM, or DVD disc in a series of pits on the disc. These pits are scanned by a laser beam. An important limitation on the amount of information that can be stored on such a disc is the width of the laser beam. Explain why this should be, and explain how using a shorter-wavelength laser allows more information to be stored on a disc of the same size.

Q36.12 With which color of light can the Hubble Space Telescope see finer detail in a distant astronomical object: red, blue, or ultraviolet? Explain your answer.

Q36.13 At the end of Section 36.4, the following statements were made about an array of N slits. Explain, using phasor diagrams, why each statement is true. (a) A minimum occurs whenever ϕ is an integer multiple of $2\pi/N$ (except when ϕ is an integer multiple of 2π, which gives a principal maximum). (b) There are $(N - 1)$ minima between each pair of principal maxima.

Q36.14 Could x-ray diffraction effects with crystals be observed by using visible light instead of x rays? Why or why not?

Q36.15 Why is a diffraction grating better than a two-slit setup for measuring wavelengths of light?

Q36.16 One sometimes sees rows of evenly spaced radio antenna towers. A student remarked that these act like diffraction gratings. What did she mean? Why would one want them to act like a diffraction grating?

Q36.17 If a hologram is made using 600 nm light and then viewed with 500 nm light, how will the images look compared to those observed when viewed with 600 nm light? Explain.

Q36.18 A hologram is made using 600 nm light and then viewed by using white light from an incandescent bulb. What will be seen? Explain.

Q36.19 Ordinary photographic film reverses black and white, in the sense that the most brightly illuminated areas become blackest upon development (hence the term *negative*). Suppose a hologram negative is viewed directly, without making a positive transparency. How will the resulting images differ from those obtained with the positive? Explain.

EXERCISES

Section 36.2 Diffraction from a Single Slit

36.1 •• Monochromatic light from a distant source is incident on a slit 0.750 mm wide. On a screen 2.00 m away, the distance from the central maximum of the diffraction pattern to the first minimum is measured to be 1.35 mm. Calculate the wavelength of the light.

36.2 • Coherent electromagnetic waves with wavelength λ pass through a narrow slit of width a. The diffraction pattern is observed on a tall screen that is 2.00 m from the slit. When $\lambda = 500$ nm, the width on the screen of the central maximum in the diffraction pattern is 8.00 mm. For the same slit and screen, what is the width of the central maximum when $\lambda = 0.125$ mm?

36.3 •• Light of wavelength 585 nm falls on a slit 0.0666 mm wide. (a) On a very large and distant screen, how many *totally* dark fringes (indicating complete cancellation) will there be, including both sides of the central bright spot? Solve this problem *without* calculating all the angles! (*Hint:* What is the largest that $\sin\theta$ can be? What does this tell you is the largest that m can be?) (b) At what angle will the dark fringe that is most distant from the central bright fringe occur?

36.4 • Light of wavelength 633 nm from a distant source is incident on a slit 0.750 mm wide, and the resulting diffraction pattern is observed on a screen 3.50 m away. What is the distance between the two dark fringes on either side of the central bright fringe?

36.5 •• Diffraction occurs for all types of waves, including sound waves. High-frequency sound from a distant source with wavelength 9.00 cm passes through a slit 12.0 cm wide. A microphone is placed 8.00 m directly in front of the center of the slit, corresponding to point O in Fig. 36.5a. The microphone is then moved in a direction perpendicular to the line from the center of the slit to point O. At what distances from O will the intensity detected by the microphone be zero?

36.6 • CP **Tsunami!** On December 26, 2004, a violent earthquake of magnitude 9.1 occurred off the coast of Sumatra. This quake triggered a huge tsunami (similar to a tidal wave) that killed more than 150,000 people. Scientists observing the wave on the open ocean measured the time between crests to be 1.0 h and the speed of the wave to be 800 km/h. Computer models of the evolution of this enormous wave showed that it bent around the continents and spread to all the oceans of the earth. When the wave reached the gaps between continents, it diffracted between them as through a slit. (a) What was the wavelength of this tsunami? (b) The distance between the southern tip of Africa and northern Antarctica is about 4500 km, while the distance between the southern end of Australia and Antarctica is about 3700 km. As an approximation, we can model this wave's behavior by using Fraunhofer diffraction. Find the smallest angle away from the central maximum for which the waves would cancel after going through each of these continental gaps.

36.7 •• CP A series of parallel linear water wave fronts are traveling directly toward the shore at 15.5 cm/s on an otherwise placid lake. A long concrete barrier that runs parallel to the shore at a distance of 3.25 m away has a hole in it. You count the wave crests and observe that 75.0 of them pass by each minute, and you also observe that no waves reach the shore at ± 62.3 cm from the point directly opposite the hole, but waves do reach the shore everywhere within this distance. (a) How wide is the hole in the barrier? (b) At what other angles do you find no waves hitting the shore?

36.8 • Monochromatic electromagnetic radiation with wavelength λ from a distant source passes through a slit. The diffraction pattern is observed on a screen 2.50 m from the slit. If the width of the central maximum is 6.00 mm, what is the slit width a if the wavelength is (a) 500 nm (visible light); (b) 50.0 μm (infrared radiation); (c) 0.500 nm (x rays)?

36.9 •• **Doorway Diffraction.** Sound of frequency 1250 Hz leaves a room through a 1.00-m-wide doorway (see Exercise 36.5). At which angles relative to the centerline perpendicular to the doorway will someone outside the room hear no sound? Use 344 m/s for the speed of sound in air and assume that the source and listener are both far enough from the doorway for Fraunhofer diffraction to apply. You can ignore effects of reflections.

36.10 • **CP** Light waves, for which the electric field is given by $E_y(x, t) = E_{max} \sin[(1.40 \times 10^7\ \text{m}^{-1})x - \omega t]$, pass through a slit and produce the first dark bands at $\pm 28.6°$ from the center of the diffraction pattern. (a) What is the frequency of this light? (b) How wide is the slit? (c) At which angles will other dark bands occur?

36.11 •• Red light of wavelength 633 nm from a helium–neon laser passes through a slit 0.350 mm wide. The diffraction pattern is observed on a screen 3.10 m away. Define the width of a bright fringe as the distance between the minima on either side. (a) What is the width of the central bright fringe? (b) What is the width of the first bright fringe on either side of the central one?

36.12 • When coherent electromagnetic waves with wavelength $\lambda = 120\ \mu$m are incident on a single slit of width a, the width of the central maximum on a tall screen 1.50 m from the slit is 90.0 cm. For the same slit and screen, for what wavelength of the incident waves is the width of the central maximum 180.0 cm, double the value when $\lambda = 120\ \mu$m?

Section 36.3 Intensity in the Single-Slit Pattern

36.13 •• Monochromatic light of wavelength 580 nm passes through a single slit and the diffraction pattern is observed on a screen. Both the source and screen are far enough from the slit for Fraunhofer diffraction to apply. (a) If the first diffraction minima are at $\pm 90.0°$, so the central maximum completely fills the screen, what is the width of the slit? (b) For the width of the slit as calculated in part (a), what is the ratio of the intensity at $\theta = 45.0°$ to the intensity at $\theta = 0$?

36.14 •• Monochromatic light of wavelength $\lambda = 620$ nm from a distant source passes through a slit 0.450 mm wide. The diffraction pattern is observed on a screen 3.00 m from the slit. In terms of the intensity I_0 at the peak of the central maximum, what is the intensity of the light at the screen the following distances from the center of the central maximum: (a) 1.00 mm; (b) 3.00 mm; (c) 5.00 mm?

36.15 •• One FM Radio station in Kuala Lumpur broadcasts at 88.1 MHz. The radio waves pass between the 452 m high Petronas Twin Towers that are 58.4 m apart along their closest walls. (a) At what horizontal angles, relative to the original direction of the waves, will a distant antenna not receive any signal from this station? (b) If the maximum intensity is 3.50 W/m^2 at the antenna, what is the intensity at $\pm 1.00°$ from the center of the central maximum at the distant antenna?

36.16 • Monochromatic light of wavelength 474 nm from a distant source passes through a slit that is 0.0340 mm wide. In the resulting diffraction pattern, the intensity at the center of the central maximum ($\theta = 0°$) is 1.04×10^{-4} W/m^2. What is the intensity at a point on the screen that corresponds to $\theta = 1.20°$?

Section 36.4 Multiple Slits

36.17 • Nearly monochromatic coherent light waves leave two rectangular slits in phase and at an angle of $\theta = 22.0°$ with the normal. When the light reaches a distant screen, the waves from the center of one slit are 344° out of phase with the waves from the center of the other slit, and the waves from the top of either slit are 172° out of phase with the waves from the bottom of that slit. (a) How is the center-to-center distance between the slits related to the width of either slit? (b) Calculate the intensity at the screen for $\theta = 22.0°$ if the intensity at $\theta = 0°$ is 0.234 W/m^2.

36.18 • Parallel rays of monochromatic light with wavelength 579 nm illuminate two identical slits and produce an interference pattern on a screen that is 75.0 cm from the slits. The centers of the slits are 0.640 mm apart and the width of each slit is 0.434 mm. If the intensity at the center of the central maximum is 3.50×10^{-4} W/m^2, what is the intensity at a point on the screen that is 0.800 mm from the center of the central maximum?

36.19 • **Number of Fringes in a Diffraction Maximum.** In Fig. 36.12c the central diffraction maximum contains exactly seven interference fringes, and in this case $d/a = 4$. (a) What must the ratio d/a be if the central maximum contains exactly five fringes? (b) In the case considered in part (a), how many fringes are contained within the first diffraction maximum on one side of the central maximum?

36.20 •• **Diffraction and Interference Combined.** Consider the interference pattern produced by two parallel slits of width a and separation d, in which $d = 3a$. The slits are illuminated by normally incident light of wavelength λ. (a) First we ignore diffraction effects due to the slit width. At what angles θ from the central maximum will the next four maxima in the two-slit interference pattern occur? Your answer will be in terms of d and λ. (b) Now we include the effects of diffraction. If the intensity at $\theta = 0°$ is I_0, what is the intensity at each of the angles in part (a)? (c) Which double-slit interference maxima are missing in the pattern? (d) Compare your results to those illustrated in Fig. 36.12c. In what ways are your results different?

36.21 •• An interference pattern is produced by light of wavelength 580 nm from a distant source incident on two identical parallel slits separated by a distance (between centers) of 0.530 mm. (a) If the slits are very narrow, what would be the angular positions of the first-order and second-order, two-slit interference maxima? (b) Let the slits have width 0.320 mm. In terms of the intensity I_0 at the center of the central maximum, what is the intensity at each of the angular positions in part (a)?

36.22 •• Laser light of wavelength 506.0 nm illuminates two identical slits, producing an interference pattern on a screen 88.0 cm from the slits. The bright bands are 1.20 cm apart, and the third bright bands on either side of the central maximum are missing in the pattern. Find the width and the separation of the two slits.

36.23 • Coherent electromagnetic waves with wavelength $\lambda = 500$ nm pass through two identical slits. The width of each slit is a, and the distance between the centers of the slits is $d = 9.00$ mm. (a) What is the smallest possible width a of the slits if the $m = 3$ maximum in the interference pattern is not present? (b) What is the next larger value of the slit width for which the $m = 3$ maximum is absent?

36.24 •• Coherent light with wavelength 200 nm passes through two identical slits. The width of each slit is a, and the distance between the centers of the slits is $d = 1.00$ mm. The $m = 5$ maximum in the two-slit interference pattern is absent, but the maxima for $m = 0$ through $m = 4$ are present. What is the ratio of the intensities for the $m = 1$ and $m = 2$ maxima in the two-slit pattern?

Section 36.5 The Diffraction Grating

36.25 • When laser light of wavelength 632.8 nm passes through a diffraction grating, the first bright spots occur at $\pm 17.8°$ from the central maximum. (a) What is the line density (in lines/cm) of this grating? (b) How many additional bright spots are there beyond the first bright spots, and at what angles do they occur?

36.26 •• Monochromatic light is at normal incidence on a plane transmission grating. The first-order maximum in the interference pattern is at an angle of 8.01°. What is the angular position of the fourth-order maximum?

36.27 • You send coherent 550 nm light through a diffraction grating that has slits of equal widths and constant separation between adjacent slits. You expect to see the fourth-order interference maximum at an angle of 66.6° with respect to the normal to the grating. However, that order is missing because 66.6° is also the angle for the third diffraction minimum (as measured from the central diffraction maximum) for each slit. (a) Find the center-to-center distance between adjacent slits. (b) Find the number of slits per mm. (c) Find the width of each slit.

36.28 • If a diffraction grating produces a third-order bright spot for red light (of wavelength 690 nm) at 65.0° from the central maximum, at what angle will the second-order bright spot be for violet light (of wavelength 405 nm)?

36.29 • Two Fraunhofer lines in the solar absorption spectrum have wavelengths of 430.790 nm and 430.774 nm. A diffraction grating has 12,800 slits. (a) What is the minimum chromatic resolving power needed to resolve these two spectral lines? (b) What is the lowest order required to resolve these two lines?

36.30 • The wavelength range of the visible spectrum is approximately 380–750 nm. White light falls at normal incidence on a diffraction grating that has 350 slits/mm. Find the angular width of the visible spectrum in (a) the first order and (b) the third order. (*Note:* An advantage of working in higher orders is the greater angular spread and better resolution. A disadvantage is the overlapping of different orders, as shown in Example 36.4.)

36.31 • (a) What is the wavelength of light that is deviated in the first order through an angle of 14.6° by a transmission grating having 5000 slits/cm? (b) What is the second-order deviation of this wavelength? Assume normal incidence.

36.32 •• **CDs and DVDs as Diffraction Gratings.** A laser beam of wavelength $\lambda = 632.8$ nm shines at normal incidence on the reflective side of a compact disc. (a) The tracks of tiny pits in which information is coded onto the CD are 1.60 μm apart. For what angles of reflection (measured from the normal) will the intensity of light be maximum? (b) On a DVD, the tracks are only 0.740 μm apart. Repeat the calculation of part (a) for the DVD.

36.33 • **Identifying Isotopes by Spectra.** Different isotopes of the same element emit light at slightly different wavelengths. A wavelength in the emission spectrum of a hydrogen atom is 656.45 nm; for deuterium, the corresponding wavelength is 656.27 nm. (a) What minimum number of slits is required to resolve these two wavelengths in second order? (b) If the grating has 500.00 slits/mm, find the angles and angular separation of these two wavelengths in the second order.

Section 36.6 X-Ray Diffraction

36.34 • If the planes of a crystal are 3.50 Å (1 Å $= 10^{-10}$ m $=$ 1 Ångstrom unit) apart, (a) what wavelength of electromagnetic waves is needed so that the first strong interference maximum in the Bragg reflection occurs when the waves strike the planes at an angle of 22.0°, and in what part of the electromagnetic spectrum do these waves lie? (See Fig. 32.4.) (b) At what other angles will strong interference maxima occur?

36.35 • X rays of wavelength 0.0850 nm are scattered from the atoms of a crystal. The second-order maximum in the Bragg reflection occurs when the angle θ in Fig. 36.22 is 21.5°. What is the spacing between adjacent atomic planes in the crystal?

36.36 • Monochromatic x rays are incident on a crystal for which the spacing of the atomic planes is 0.440 nm. The first-order maximum in the Bragg reflection occurs when the incident and reflected x rays make an angle of 39.4° with the crystal planes. What is the wavelength of the x rays?

Section 36.7 Circular Apertures and Resolving Power

36.37 •• Monochromatic light with wavelength 620 nm passes through a circular aperture with diameter 7.4 μm. The resulting diffraction pattern is observed on a screen that is 4.5 m from the aperture. What is the diameter of the Airy disk on the screen?

36.38 •• Monochromatic light with wavelength 490 nm passes through a circular aperture, and a diffraction pattern is observed on a screen that is 1.20 m from the aperture. If the distance on the screen between the first and second dark rings is 1.65 mm, what is the diameter of the aperture?

36.39 • Two satellites at an altitude of 1100 km are separated by 26 km. If they broadcast 3.7 cm microwaves, what minimum receiving-dish diameter is needed to resolve (by Rayleigh's criterion) the two transmissions?

36.40 • **BIO** If you can read the bottom row of your doctor's eye chart, your eye has a resolving power of 1 arcminute, equal to $\frac{1}{60}$ degree. If this resolving power is diffraction limited, to what effective diameter of your eye's optical system does this correspond? Use Rayleigh's criterion and assume $\lambda = 550$ nm.

36.41 •• The VLBA (Very Long Baseline Array) in New Mexico uses a number of individual radio telescopes to make one unit having an equivalent diameter of about 8000 km. When this radio telescope is focusing radio waves of wavelength 2.0 cm, what would have to be the diameter of the mirror of a visible-light telescope focusing light of wavelength 550 nm so that the visible-light telescope has the same resolution as the radio telescope?

36.42 • **Photography.** A wildlife photographer uses a moderate telephoto lens of focal length 135 mm and maximum aperture $f/4.00$ to photograph a bear that is 11.5 m away. Assume the wavelength is 550 nm. (a) What is the width of the smallest feature on the bear that this lens can resolve if it is opened to its maximum aperture? (b) If, to gain depth of field, the photographer stops the lens down to $f/22.0$, what would be the width of the smallest resolvable feature on the bear?

36.43 •• **Hubble Versus FAST.** The Hubble Space Telescope has an aperture of 2.4 m and focuses visible light (380–750 nm). The FAST (Five hundred meter Aperture Spherical Telescope) radio telescope in Guizhou province, China is 500 m in diameter (it is built in a mountain valley) and focuses radio waves of wavelength 22 cm. (a) Under optimal viewing conditions, what is the smallest crater that each of these telescopes could resolve on our moon? (b) If the Hubble Space Telescope were to be converted to surveillance use, what is the highest orbit above the surface of the earth it could have and still be able to resolve an A4 size (210 by 297 mm) sheet of paper (not the writing, just the page) on the ground? Assume optimal viewing conditions, so that the resolution is diffraction limited.

36.44 • **Observing Jupiter.** You are asked to design a space telescope for earth orbit. When Jupiter is 5.93×10^8 km away (its closest approach to the earth), the telescope is to resolve, by Rayleigh's criterion, features on Jupiter that are 250 km apart. What minimum-diameter mirror is required? Assume a wavelength of 500 nm.

PROBLEMS

36.45 •• **BIO** **Thickness of Human Hair.** Although we have discussed single-slit diffraction only for a slit, a similar result holds when light bends around a straight, thin object, such as a strand of hair. In that case, a is the width of the strand. From actual laboratory measurements on a human hair, it was found that when a beam of light of wavelength 632.8 nm was shone on a single strand of hair, and the diffracted light was viewed on a screen 1.25 m away, the first dark fringes on either side of the central bright spot were 5.22 cm apart. How thick was this strand of hair?

36.46 •• **CP** A loudspeaker with a diaphragm that vibrates at 1150 Hz is traveling at 95.0 m/s directly toward a pair of holes in a very large wall. The speed of sound in the region is 344 m/s. Far from the wall, you observe that the sound coming through the openings first cancels at $\pm 12.7°$ with respect to the direction in which the speaker is moving. (a) How far apart are the two openings? (b) At what angles would the sound first cancel if the source stopped moving?

36.47 ••• Laser light of wavelength 632.8 nm falls normally on a slit that is 0.0290 mm wide. The transmitted light is viewed on a distant screen where the intensity at the center of the central bright fringe is 8.00 W/m^2. (a) Find the maximum number of totally dark fringes on the screen, assuming the screen is large enough to show them all. (b) At what angle does the dark fringe that is most distant from the center occur? (c) What is the maximum intensity of the bright fringe that occurs immediately before the dark fringe in part (b)? Approximate the angle at which this fringe occurs by assuming it is midway between the angles to the dark fringes on either side of it.

36.48 • **Grating Design.** Your boss asks you to design a diffraction grating that will disperse the first-order visible spectrum through an angular range of 27.0°. (See Example 36.4 in Section 36.5.) (a) What must be the number of slits per centimeter for this grating? (b) At what angles will the first-order visible spectrum begin and end?

36.49 • **Measuring Refractive Index.** A thin slit illuminated by light of frequency f produces its first dark band at $\pm 38.2°$ in air. When the entire apparatus (slit, screen, and space in between) is immersed in an unknown transparent liquid, the slit's first dark bands occur instead at $\pm 21.6°$. Find the refractive index of the liquid.

36.50 •• **Underwater Photography.** An underwater camera has a lens with focal length in air of 35.0 mm and a maximum aperture of $f/2.80$. The film it uses has an emulsion that is sensitive to light of frequency 6.00×10^{14} Hz. If the photographer takes a picture of an object 2.75 m in front of the camera with the lens wide open, what is the width of the smallest resolvable detail on the subject if the object is (a) a fish underwater with the camera in the water and (b) a person on the beach with the camera out of the water?

36.51 ••• **CALC** The intensity of light in the Fraunhofer diffraction pattern of a single slit is given by Eq. (36.5). Let $\gamma = \beta/2$. (a) Show that the equation for the values of γ at which I is a maximum is $\tan\gamma = \gamma$. (b) Determine the two smallest positive values of γ that are solutions of this equation. (*Hint:* You can use a trial-and-error procedure. Guess a value of γ and adjust your guess to bring $\tan\gamma$ closer to γ. A graphical solution of the equation is very helpful in locating the solutions approximately, to get good initial guesses.) (c) What are the positive values of γ for the first, second, and third minima on one side of the central maximum? Are the γ values in part (b) precisely halfway between the γ values for adjacent minima? (d) If $a = 12\lambda$, what are the angles θ (in degrees) that locate the first minimum, the first maximum beyond the central maximum, and the second minimum?

36.52 •• A slit 0.400 mm wide is illuminated by parallel rays of light that have a wavelength of 600 nm. The diffraction pattern is observed on a screen that is 1.40 m from the slit. The intensity at the center of the central maximum ($\theta = 0°$) is I_0. (a) What is the distance on the screen from the center of the central maximum to the first minimum? (b) What is the distance on the screen from the center of the central maximum to the point where the intensity has fallen to $I_0/2$?

36.53 •• **CP CALC** In a large vacuum chamber, monochromatic laser light passes through a narrow slit in a thin aluminum plate and forms a diffraction pattern on a screen that is 0.620 m from the slit. When the aluminum plate has a temperature of 20.0°C, the width of the central maximum in the diffraction pattern is 2.06 mm. What is the change in the width of the central maximum when the temperature of the plate is raised to 420.0°C? Does the width of the central diffraction maximum increase or decrease when the temperature is increased?

36.54 •• **CP** In a laboratory, light from a particular spectrum line of helium passes through a diffraction grating and the second-order maximum is at 12.2° from the center of the central bright fringe. The same grating is then used for light from a distant galaxy that is moving away from the earth with a speed of 2.61×10^7 m/s. For the light from the galaxy, what is the angular location of the second-order maximum for the same spectral line as was observed in the lab? (See Section 16.8.)

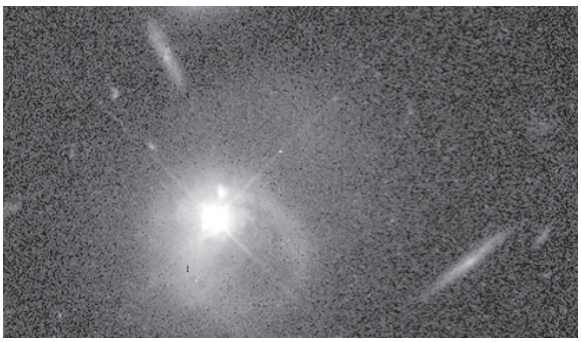
36.55 • What is the longest wavelength that can be observed in the third order for a transmission grating having 9200 slits/cm? Assume normal incidence.

36.56 •• Humans perceive sound with frequencies in the range 20 to 20,000 Hz. Speech lies in the middle of this range, 400 to 1000 Hz, while bells, sirens, and the knock of silverware on plates all extend above the range for speech. Musical instruments that keep rhythm, such as drums and bass guitars, supply sounds with lower frequencies. Doorways act as natural filters, so noises sound different when they come from around corners. (a) Estimate the width of a typical doorway in your house. (b) Determine the sound frequency that corresponds to a wavelength comparable to the width of a doorway. (Such waves freely diffract around the corner of the door.) (c) Determine the frequency of sound that would diffract only $\pm 20°$ from directly forward when passing through a doorway. (d) Based on your estimates, does a doorway act as a noticeable filter for sounds we can hear? (e) If so, which frequency range (low, middle, high) is harder to hear from around the corner of a doorway? (f) Experiment to see if you can notice this effect. It is more readily perceived from outside a house as you listen to sounds from inside. Why is this so?

36.57 • A diffraction grating has 650 slits/mm. What is the highest order that contains the entire visible spectrum? (The wavelength range of the visible spectrum is approximately 380–750 nm.)

36.58 •• *Quasars,* an abbreviation for *quasi-stellar radio sources,* are distant objects that look like stars through a telescope but that emit far more electromagnetic radiation than an entire normal galaxy of stars. An example is the bright object below and to the left of center in **Fig. P36.58**; the other elongated objects in this image are normal galaxies. The leading model for the structure of a quasar is a galaxy with a supermassive black hole at its center. In this model, the radiation is emitted by interstellar gas and dust within the galaxy as this material falls toward the black hole. The radiation is thought to emanate from a region just a few light-years in diameter. (The diffuse glow surrounding the bright quasar shown in Fig. P36.58 is thought to be this quasar's host galaxy.) To investigate this model of quasars and to study other exotic astronomical objects, the Russian Space Agency has placed a radio telescope in a large orbit around the earth. When this telescope is 77,000 km from earth and the signals it receives are combined with signals from the ground-based telescopes of the VLBA, the resolution is that of a single radio telescope 77,000 km in diameter. What is the size of the smallest detail that this arrangement can resolve in quasar 3C 405, which is 7.2×10^8 light-years from earth, using radio waves at a frequency of 1665 MHz? (*Hint:* Use Rayleigh's criterion.) Give your answer in light-years and in kilometers.

Figure **P36.58**

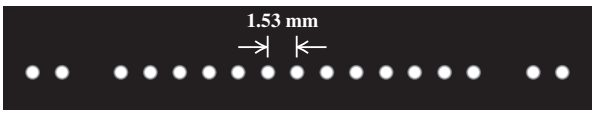


36.59 •• In the 1920s Clinton Davisson and Lester Germer accidentally observed diffraction when electrons with 54 eV of energy were scattered off crystalline nickel. The diffraction peak occurred when the angle between the incident beam and the scattered beam was 50°. (a) What is the corresponding angle θ relevant for Eq. (36.16)? (b) The planes in crystalline nickel are separated by 0.091 nm, as determined by x-ray scattering experiments. According to the Bragg condition, what wavelength do the electrons in these experiments have? (c) Given the mass of an electron as 9.11×10^{-31} kg, what is the corresponding classical speed v_{cl} of the diffracted electrons? (d) Assuming the electrons correspond to a wave with speed v_{cl} and wavelength λ, what is the frequency f of the diffracted waves? (e) Quantum mechanics postulates that the energy E and the frequency f of a particle are related by $E = hf$, where h is known as Planck's constant. Estimate h from these observations. (f) Our analysis has a small flaw: The relevant wave velocity, known as a quantum phase velocity, is *half* the classical particle velocity, for reasons explained by deeper aspects of quantum physics. Re-estimate the value of h using this modification. (g) The established value of Planck's constant is 6.6×10^{-34} J·s. Does this agree with your estimate?

36.60 •• BIO **Resolution of the Eye.** The maximum resolution of the eye depends on the diameter of the opening of the pupil (a diffraction effect) and the size of the retinal cells. The size of the retinal cells (about 5.0 μm in diameter) limits the size of an object at the near point (25 cm) of the eye to a height of about 50 μm. (To get a reasonable estimate without having to go through complicated calculations, we shall ignore the effect of the fluid in the eye.) (a) Given that the diameter of the human pupil is about 2.0 mm, does the Rayleigh criterion allow us to resolve a 50-μm-tall object at 25 cm from the eye with light of wavelength 550 nm? (b) According to the Rayleigh criterion, what is the shortest object we could resolve at the 25 cm near point with light of wavelength 550 nm? (c) What angle would the object in part (b) subtend at the eye? Express your answer in minutes (60 min = 1°), and compare it with the experimental value of about 1 min. (d) Which effect is more important in limiting the resolution of our eyes: diffraction or the size of the retinal cells?

36.61 •• DATA While researching the use of laser pointers, you conduct a diffraction experiment with two thin parallel slits. Your result is the pattern of closely spaced bright and dark fringes shown in **Fig. P36.61**. (Only the central portion of the pattern is shown.) You measure that the bright spots are equally spaced at 1.53 mm center to center (except for the missing spots) on a screen that is 2.50 m from the slits. The light source was a helium–neon laser producing a wavelength of 632.8 nm. (a) How far apart are the two slits? (b) How wide is each one?

Figure P36.61



36.62 •• DATA Your physics study partner tells you that the width of the central bright band in a single-slit diffraction pattern is inversely proportional to the width of the slit. This means that the width of the central maximum increases when the width of the slit decreases. The claim seems counterintuitive to you, so you make measurements to test it. You shine monochromatic laser light with wavelength λ onto a very narrow slit of width a and measure the width w of the central maximum in the diffraction pattern that is produced on a screen 1.50 m from the slit. (By "width," you mean the distance on the screen between the two minima on either side of the central maximum.) Your measurements are given in the table.

a (μm)	0.78	0.91	1.04	1.82	3.12	5.20	7.80	10.40	15.60
w (m)	2.68	2.09	1.73	0.89	0.51	0.30	0.20	0.15	0.10

(a) If w is inversely proportional to a, then the product aw is constant, independent of a. For the data in the table, graph aw versus a. Explain why aw is not constant for smaller values of a. (b) Use your graph in part (a) to calculate the wavelength λ of the laser light. (c) What is the angular position of the first minimum in the diffraction pattern for (i) $a = 0.78$ μm and (ii) $a = 15.60$ μm?

36.63 •• DATA At the metal fabrication company where you work, you are asked to measure the diameter D of a very small circular hole in a thin, vertical metal plate. To do so, you pass coherent monochromatic light with wavelength 562 nm through the hole and observe the diffraction pattern on a screen that is a distance x from the hole. You measure the radius r of the first dark ring in the diffraction pattern (see Fig. 36.26). You make the measurements for four values of x. Your results are given in the table.

x (m)	1.00	1.50	2.00	2.50
r (cm)	5.6	8.5	11.6	14.1

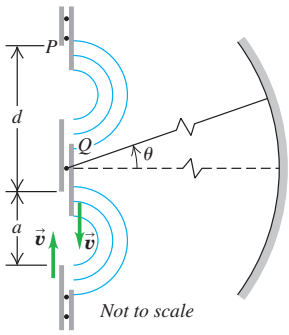
(a) Use each set of measurements to calculate D. Because the measurements contain some error, calculate the average of the four values of D and take that to be your reported result. (b) For $x = 1.00$ m, what are the radii of the second and third dark rings in the diffraction pattern?

36.64 •• A glass sheet is covered by a very thin opaque coating. In the middle of this sheet there is a thin scratch 0.00125 mm thick. The sheet is totally immersed beneath the surface of a liquid. Parallel rays of monochromatic coherent light with wavelength 612 nm in air strike the sheet perpendicular to its surface and pass through the scratch. A screen is placed in the liquid a distance of 30.0 cm away from the sheet and parallel to it. You observe that the first dark fringes on either side of the central bright fringe on the screen are 22.4 cm apart. What is the refractive index of the liquid?

CHALLENGE PROBLEMS

36.65 ••• An opaque barrier has an inner membrane and an outer membrane that slide past each other, as shown in **Fig. P36.65**. Each membrane includes parallel slits of width a separated by a distance d. A screen forms a circular arc subtending 60° at the fixed midpoint between the slits. A green 532 nm laser impinges on the slits from the left. The outer membrane moves upward with speed v while the inner membrane moves downward with the same speed, propelled by nanomotors. At time $t = 0$, point P on the outer membrane is adjacent to point Q on the inner membrane so that the effective aperture width is zero. The aperture is fully closed again at $t = 3.00$ s. (a) At $t = 1.00$ s, there are 19 evenly spaced bright spots on the screen, each of approximately the same intensity. At the edges of the screen the first diffraction minimum and a two-slit interference maximum coincide. What is the slit distance d? (Note: The screen does not encompass the entire diffraction pattern.) (b) What is the speed v? (c) What is the maximum aperture width a? (d) At a certain time, the outermost spots (the $m = \pm 9$ spots) disappear. What is that time? (e) At $t = 1.50$ s, what is the intensity of the $m = \pm 1$ spots in terms of the $m = 0$ central spot? (f) What are the angular positions of these spots?

Figure **P36.65**



36.66 ••• **Intensity Pattern of N Slits.** (a) Consider an arrangement of N slits with a distance d between adjacent slits. The slits emit coherently and in phase at wavelength λ. Show that at a time t, the electric field at a distant point P is

$$\begin{aligned} E_P(t) &= E_0\cos(kR - \omega t) + E_0\cos(kR - \omega t + \phi) \\ &\quad + E_0\cos(kR - \omega t + 2\phi) + \cdots \\ &\quad + E_0\cos(kR - \omega t + (N-1)\phi) \end{aligned}$$

where E_0 is the amplitude at P of the electric field due to an individual slit, $\phi = (2\pi d\sin\theta)/\lambda$, θ is the angle of the rays reaching P (as measured from the perpendicular bisector of the slit arrangement), and R is the distance from P to the most distant slit. In this problem, assume that R is much larger than d. (b) To carry out the sum in part (a), it is convenient to use the complex-number relationship $e^{iz} = \cos z + i\sin z$, where $i = \sqrt{-1}$. In this expression, $\cos z$ is the *real part* of the complex number e^{iz}, and $\sin z$ is its *imaginary part*. Show that the electric field $E_P(t)$ is equal to the real part of the complex quantity

$$\sum_{n=0}^{N-1} E_0 e^{i(kR-\omega t+n\phi)}$$

(c) Using the properties of the exponential function that $e^A e^B = e^{(A+B)}$ and $(e^A)^n = e^{nA}$, show that the sum in part (b) can be written as

$$\begin{aligned} &E_0\left(\frac{e^{iN\phi} - 1}{e^{i\phi} - 1}\right)e^{i(kR-\omega t)} \\ &= E_0\left(\frac{e^{iN\phi/2} - e^{-iN\phi/2}}{e^{i\phi/2} - e^{-i\phi/2}}\right)e^{i[kR-\omega t+(N-1)\phi/2]} \end{aligned}$$

Then, using the relationship $e^{iz} = \cos z + i\sin z$, show that the (real) electric field at point P is

$$E_P(t) = \left[E_0\frac{\sin(N\phi/2)}{\sin(\phi/2)}\right]\cos[kR - \omega t + (N-1)\phi/2]$$

The quantity in the first square brackets in this expression is the amplitude of the electric field at P. (d) Use the result for the electric-field amplitude in part (c) to show that the intensity at an angle θ is

$$I = I_0\left[\frac{\sin(N\phi/2)}{\sin(\phi/2)}\right]^2$$

where I_0 is the maximum intensity for an individual slit. (e) Check the result in part (d) for the case $N = 2$. It will help to recall that $\sin 2A = 2\sin A\cos A$. Explain why your result differs from Eq. (35.10), the expression for the intensity in two-source interference, by a factor of 4. (*Hint:* Is I_0 defined in the same way in both expressions?)

36.67 ••• CALC **Intensity Pattern of N Slits, Continued.** Part (d) of Challenge Problem 36.66 gives an expression for the intensity in the interference pattern of N identical slits. Use this result to verify the following statements. (a) The maximum intensity in the pattern is $N^2 I_0$. (b) The principal maximum at the center of the pattern extends from $\phi = -2\pi/N$ to $\phi = 2\pi/N$, so its width is inversely proportional to $1/N$. (c) A minimum occurs whenever ϕ is an integer multiple of $2\pi/N$, except when ϕ is an integer multiple of 2π (which gives a principal maximum). (d) There are $(N-1)$ minima between each pair of principal maxima. (e) Halfway between two principal maxima, the intensity can be no greater than I_0; that is, it can be no greater than $1/N^2$ times the intensity at a principal maximum.

36.68 ••• CALC It is possible to calculate the intensity in the single-slit Fraunhofer diffraction pattern *without* using the phasor method of Section 36.3. Let y' represent the position of a point within the slit of width a in Fig. 36.5a, with $y' = 0$ at the center of the slit so that the slit extends from $y' = -a/2$ to $y' = a/2$. We imagine dividing the slit up into infinitesimal strips of width dy', each of which acts as a source of secondary wavelets. (a) The amplitude of the total wave at the point O on the distant screen in Fig. 36.5a is E_0. Explain why the amplitude of the wavelet from each infinitesimal strip within the slit is $E_0(dy'/a)$, so that the electric field of the wavelet a distance x from the infinitesimal strip is $dE = E_0(dy'/a)\sin(kx - \omega t)$. (b) Explain why the wavelet from each strip as detected at point P in Fig. 36.5a can be expressed as

$$dE = E_0\frac{dy'}{a}\sin[k(D - y'\sin\theta) - \omega t]$$

where D is the distance from the center of the slit to point P and $k = 2\pi/\lambda$. (c) By integrating the contributions dE from all parts of the slit, show that the total wave detected at point P is

$$\begin{aligned} E &= E_0\sin(kD - \omega t)\frac{\sin[ka(\sin\theta)/2]}{ka(\sin\theta)/2} \\ &= E_0\sin(kD - \omega t)\frac{\sin[\pi a(\sin\theta)/\lambda]}{\pi a(\sin\theta)/\lambda} \end{aligned}$$

(The trigonometric identities in Appendix D will be useful.) Show that at $\theta = 0°$, corresponding to point O in Fig. 36.5a, the wave is $E = E_0\sin(kD - \omega t)$ and has amplitude E_0, as stated in part (a). (d) Use the result of part (c) to show that if the intensity at point O is I_0, then the intensity at a point P is given by Eq. (36.7).

MCAT-STYLE PASSAGE PROBLEMS

Bragg Reflection on a Different Scale. A *colloid* consists of particles of one type of substance dispersed in another substance. Suspensions of electrically charged microspheres (microscopic spheres, such as polystyrene) in a liquid such as water can form a colloidal crystal when the microspheres arrange themselves in a regular repeating pattern under the influence of the electrostatic force. Colloidal crystals can selectively manipulate different wavelengths of visible light. Just as we can study crystalline solids by using Bragg reflection of x rays, we can study colloidal crystals through Bragg scattering of visible light from the regular arrangement of charged microspheres. Because the light is traveling through a liquid when it experiences the path differences that lead to constructive interference, it is the wavelength in the liquid that determines the angles at which Bragg reflections are seen. In one experiment, laser light with a wavelength in vacuum of 650 nm is passed through a sample of charged polystyrene spheres in water. A strong interference maximum is then observed when the incident and reflected beams make an angle of 39° with the colloidal crystal planes.

36.69 Why is visible light, which has much longer wavelengths than x rays do, used for Bragg reflection experiments on colloidal crystals? (a) The microspheres are suspended in a liquid, and it is more difficult for x rays to penetrate liquid than it is for visible light. (b) The irregular spacing of the microspheres allows the longer-wavelength visible light to produce more destructive interference than can x rays. (c) The microspheres are much larger than atoms in a crystalline solid, and in order to get interference maxima at reasonably large angles, the wavelength must be much longer than the size of the individual scatterers. (d) The microspheres are spaced more widely than atoms in a crystalline solid, and in order to get interference maxima at reasonably large angles, the wavelength must be comparable to the spacing between scattering planes.

36.70 What plane spacing in the colloidal crystal could produce the maximum in this experiment? (a) 390 nm; (b) 520 nm; (c) 650 nm; (d) 780 nm.

36.71 When the light is passed through the bottom of the sample container, the interference maximum is observed to be at 41°; when it is passed through the top, the corresponding maximum is at 37°. What is the best explanation for this observation? (a) The microspheres are more tightly packed at the bottom, because they tend to settle in the suspension. (b) The microspheres are more tightly packed at the top, because they tend to float to the top of the suspension. (c) The increased pressure at the bottom makes the microspheres smaller there. (d) The maximum at the bottom corresponds to $m = 2$, whereas the maximum at the top corresponds to $m = 1$.

ANSWERS

Chapter Opening Question ?

(i) For an optical system that uses a lens, the ability to resolve fine details—its resolving power, or resolution—improves as the lens diameter D increases (Section 36.7). Each miniature lens in a fly's eye produces its own image, so these images have very poor resolution, compared to those produced by a human eye, because the lens is so small. However, a fly's eye is much better than a human eye at detecting movement.

Key Example √ARIATION Problems

VP36.3.1 (a) 2.30 m (b) 22.8 mm
VP36.3.2 (a) 673 nm (b) 0.349°
VP36.3.3 (a) 0.696° (b) $(3.11 \times 10^{-3})I_0$
VP36.3.4 (a) 0.178° (b) $0.405I_0$
VP36.5.1 (a) 398 nm (a) 19.1° for $m = 1$, 79.8° for $m = 3$
VP36.5.2 (a) 651 slits/mm (b) 54.4° for $m = 2$; there is no $m = 3$ bright region
VP36.5.3 (a) 22.1° (b) 48.7°
VP36.5.4 (a) 0.186 nm (b) 0.105 nm
VP36.6.1 (a) 2.50×10^2 m (b) 1.11×10^{-3} rad $= 0.0640°$
VP36.6.2 (a) 466 m (b) 0.00268 mm
VP36.6.3 (a) 8.46 mm (b) $f/16.0$
VP36.6.4 (a) 150 ly (b) 74 ly

Bridging Problem

1.501×10^7 m/s or 5.00% of c; away from us